

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{CRs+1}{Cs} I_2(s)$$

$$V_e(s) = \left[(Ls+2R) \frac{CLs^2+3CRs+1}{CLs^2+CRs} - (Ls+R) \right] I_2(s)$$

$$= \frac{3CLR s^2 + (5CR^2+L)s + 2R}{CLs^2+CRs} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CRs+1}{Cs} \frac{3CLR s^2 + (5CR^2+L)s + 2R}{CLs^2+CRs} I_2(s)$$

$$= \frac{(CRs+1)(Ls+R)}{3CLR s^2 + (5CR^2+L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (CR^2+L)s + R}{3CLR s^2 + (5CR^2+L)s + 2R}$$

Estabilidad en lazo abierto

$$R = 330 \, \Omega$$

$$L = 470 \, \mu H$$

$$C = 470 \, \mu F$$

$$3CLRs^2 + (5CR^2 + L)s + 2R = 0$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 3CLR = 3(470 \, \mu F)(470 \, \mu F)(330) = 2.1869 \times 10^{-4}$$

$$b = 5CR^2 + L = 255.91547$$

$$c = 2R = 660$$

$$\lambda_1 = -2.578$$

$$\lambda_2 = -1170217.687 \rightarrow 1.1702 \times 10^6$$

Error en estado estacionario $1/2 = 50\%$

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) = \left(1 - \frac{V_s(s)}{V_e(s)}\right)$$

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) = \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R}\right)$$

Todos los valores con s se vanen 0

$$3CLRs^2 + (5CR^2 + L)s + 2R$$

$\therefore$

$$e(s) = \lim_{s \rightarrow 0} s \left(\frac{1}{s} \left(1 - \frac{\cancel{CLRs^2}^0 + \cancel{(CR^2 + L)s}^0 + R}{\cancel{3CLRs^2}^0 + \cancel{(5CR^2 + L)s}^0 + 2R} \right) \right) = \frac{R}{2R} = \frac{1}{2} = 0.5$$

$$e(s) = 0.5 \text{ V}$$

50%

V_{cc}

Error

V
0.5

0.5V

0

transitorio

t

estado estacionario

1.9395

